- Over an algebraically closed field, a finite dim algebra A/k:

A has finite global dim $\Leftrightarrow$ A is smooth.

In this lecture series, assume all base fields are algebraically closed.

- Let $\mathcal{C}$ be a small DG category. $Perf(\mathcal{C})$ is constructed as follows:

(1) Take all finite semifree DG $\mathcal{C}$-modules in $Mod_{dg}(\mathcal{C})$.

(2) Take all objects in $Mod_{dg}(\mathcal{C})$ that are isomorphic to direct summands of the above objects when passing to $D(\mathcal{C})$. Denote this category by $Mod_{dg}(\mathcal{C})^c$.

(3) Define $Perf(\mathcal{C}) := Mod_{dg}(\mathcal{C})^c / Ac(\mathcal{C})$.

We have: $H^0(Perf(\mathcal{C})) = Perf(\mathcal{C})$.

$\mathcal{C}$ = small DG category. $\mathcal{C}^e := \mathcal{C}^{op} \otimes_k \mathcal{C}$

Diagonal bimodule $\Delta_{\mathcal{C}} \in Mod_{dg}(\mathcal{C}^e)$.

$$\Delta_{\mathcal{C}} : \mathcal{C}^e \longrightarrow Mod_{dg}(k)$$

$$X \otimes Y \longmapsto Hom_{\mathcal{C}}(X, Y)$$

Def: Hochschild homology of $\mathcal{C}$ is:

$$HH_n(\mathcal{C}) = H^{-n}\left(\Delta_{\mathcal{C}} \overset{L}{\otimes}_{\mathcal{C}^e} \Delta_{\mathcal{C}}\right)$$

Hochschild cohomology of $\mathcal{C}$ is:

$$HH^n(\mathcal{C}) = H^n\left(R Hom_{\mathcal{C}^e}(\Delta_{\mathcal{C}}, \Delta_{\mathcal{C}})\right).$$

Thm [Keller: On DG categories '06]

There are iso:

$$HH_*(\mathcal{C}) = HH_*(Perf(\mathcal{C}))$$

$$HH^*(\mathcal{C}) = HH^*(Perf(\mathcal{C}))$$

Thm [Keller: Invariance & localization ... 1998]

If $\mathcal{A} \to \mathcal{B} \to \mathcal{C}$ is an exact seq of small DG categories (e.g. $\mathcal{C} = \mathcal{A}/\mathcal{B}$), then we have exact triangle of Hochschild complexes: $HH_\bullet(\mathcal{A}) \to HH_\bullet(\mathcal{B}) \to HH_\bullet(\mathcal{C}) \to HH_\bullet(\mathcal{A})[1]$

Smoothness, singularity categories.

Def: A small DG category $\mathcal{C}$ is <u>smooth</u>, if $\Delta_{\mathcal{C}} \in Perf(\mathcal{C}^e)$.

A small DG cat $\mathcal{C}$ is <u>proper</u>, if in $Perf(\mathcal{C})$, $\sum_{i \in \mathbb{Z}} \dim H^i(Hom_{\mathcal{C}}(X,Y)) < \infty$ for every $X, Y \in Perf(\mathcal{C})$.

If $\mathcal{C}$ is a finite dim alg $\Rightarrow$ properness.

Let A be a finite dim algebra. Let

$\mathcal{C}_{dg}^b(mod(A)) =$ DG cat of <u>bounded</u> complexes of objects in $mod(A)$.

$$\mathcal{D}^b(mod(A)) = \mathcal{C}_{dg}^b(mod(A)) / A^c.$$

$$H^0(\mathcal{D}^b(mod(A)) = D^b(mod(A)).$$

Def: The DG singularity category of A is

$$sg(A) := \mathcal{D}^b(mod(A)) / Perf(A).$$

Rmk: ① $H^0(sg(A)) = D_{sg}(A) := D^b(mod A)/Perf(A)$.

② we have exact seq of DG categories:

$$Perf(A) \longrightarrow \mathcal{D}^b(mod(A)) \longrightarrow sg(A).$$

(2) A smooth $\Longleftrightarrow$ A finite gl.dim $\Longleftrightarrow$ $D_{sg}(A)=0$.

$HH_*(A)$ and $HH_*(sg(A))$.

Start with $Perf(A) \to D^b(mod(A)) \to sg(A)$.

We have LES: $HH_*(A)$

$$HH_*(Perf(A)) \to HH_*(D^b(mod(A))) \to HH_*(sg(A)) \to \cdots$$

Recall: $HH_*(Perf(A)) \cong HH_*(A)$

By Morita equivalence, we assume A is basic.

$R = A/J$, then R generate $D^b(mod(A))$.

$A^! := R\mathrm{Hom}_A(R,R)$ is a DG model of $D^b(mod A)$.

i.e. $D^b(mod(A)) \cong Perf(A^!)$

$$HH_*(D^b(mod(A))) \cong HH_*(Perf(A^!)) = HH_*(A^!)$$

[Van den Bergh Calabi-Yau alg & superpotentials '15]

[Wang-Arunachalam-Keller 2022]

$HH_n(A^!) \cong DHH_{-n}(A)$, $D(-) = \mathrm{Hom}_k(-,k)$

Since A is an ordinary algebra, we have

$$HH_{<0}(A)=0 \Rightarrow HH_{>0}(A^!)=0.$$

Thm [Wang-Arunachalam-Keller '22]

A finite dim ordinary basic algebra:

For $n \geq 2$: $HH_n(sg(A)) \cong HH_{n-1}(A)$,

when $n=1$: $HH_1(sg(A)) \cong \ker(HH_0(A) \to DHH_0(A))$

Rmk: If $HH_n(A) = 0$ for $n \geq 1$ $\Leftrightarrow$

$HH_n(sg(A)) = 0$ for $n \geq 2$.

If $HH_*(sg(A)) = 0$, then above condition is satisfied.

If Hanś: finite dim ordinary alg A, s.t. (conj is wrong)

$HH_{\geq 2}(sg(A)) = 0$, but A is not smooth

$\Uparrow$ $\Downarrow$

$[HH_*(sg(A)) = 0 \not\Rightarrow D_{sg}(A) \neq 0]$

We are asking for phenomenon: $?: ? \neq 0$, $HH_*(?) = 0$

phantom category.

Phantom categories.

Let $\mathcal{C}$ = pretriangulated DG cat. Let

$A, B \subseteq \mathcal{C}$ full pretriang sub.

we write $\mathcal{C} = \langle A, B \rangle$ if

- $\mathrm{Hom}_{H^0(\mathcal{C})}(B, A) = 0$, for $\forall A \in \mathcal{A}, B \in \mathcal{B}$.

and

- $\mathcal{A}, \mathcal{B}$ generate $H^0(\mathcal{C})$ by taking cones, shifts.

$\mathcal{A}$ is left admissible, $\mathcal{B}$ is right admissible,

$\mathcal{A} \simeq_{\text{Morita}} \mathcal{C}/\mathcal{B}$, $\mathcal{B} \simeq_{\text{Morita}} \mathcal{C}/\mathcal{A}$.

In particular:

Lemma: $HH_*(\mathcal{C}) = HH_*(\mathcal{A}) \oplus HH_*(\mathcal{B})$.

Example: $A = Q/I$, Q is a directed quiver.

Vertices: $1 < 2 < \cdots < n$.

$\mathcal{D}^{perf}(A) = \langle P_1, P_2, \cdots, P_n \rangle$.

Def: A smooth and proper DG category $\mathcal{C}$ is a phantom category, if

$\mathcal{C} \neq 0$, $HH_*(\mathcal{C}) = 0$, $K_0(\mathcal{C}) = 0$.

Quasi-phantom categories:

[Bühling – Graf von Bothmer – Sosna, 2013]

[Alexeev – Orlov, 2013]

Phantom categories:

[Gorchinskiy–Orlov 2013],

[Bühling–Graf von Bothmer–Katzarkov–Sosna 2015]

More recently:

[Krah 2024] will be main ingredient for our purpose.

Lemma 12: Assume $\mathcal{C}$ is smooth proper DG category with $\langle E_1, E_2, \cdots, E_n \rangle$ (i.e. $\mathrm{Ext}^*(E_i, E_i) = k$) full exceptional collection.

Suppose $\mathcal{T} = \langle F_1, F_2, \cdots, F_n \rangle$, and $\mathcal{T} \subsetneq \mathcal{C}$. Then $\mathcal{C} = \langle \mathcal{T}, \mathcal{P} \rangle$, and $\mathcal{P}$ is a phantom category.

Proof: $\mathcal{P} \neq 0$ since $\mathcal{T} \neq \mathcal{C}$.

$$HH_*(\mathcal{C}) = HH_*(\mathcal{T}) \oplus HH_*(\mathcal{P})$$

since $\mathcal{C} = \langle E_1, \cdots, E_n \rangle$, $HH_*(\mathcal{C}) = k^{\oplus n}$

$\mathcal{T} = \langle F_1, \cdots, F_n \rangle$, $HH_*(\mathcal{T}) = k^{\oplus n}$

Hence $HH_*(\mathcal{P}) = 0$, similar for $K_0(\mathcal{P}) = 0$.

Theorem [Krah 2024]

Let $X_{10} = Bl_{10}(\mathbb{P}^2)$, blow up 10 general pts.

(1) $\mathrm{Perf}(X_{10})$ admits a full strong exceptional collection of length 13.

(2) There is a exceptional collection of 13 line bundles $\langle L_1, \dots, L_{13} \rangle$, such that

$$\langle L_1, \dots, L_{13} \rangle \subsetneq \mathrm{Perf}(X_{10}).$$

(3) So residual $\mathcal{P}$ is a phantom.

Remark: [L-Shen '26]: phantom on 3-vertex directed DG quiver.